

Kenji Tojo

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About

Kenji Tojo is a postdoctoral researcher at ETH Zürich, working with Bernd Bickel. He received his Ph.D. from the University of Tokyo, advised by Nobuyuki Umetani, where he specialized in computer graphics and computational design. He has published multiple first-author papers at top venues, including SIGGRAPH, and his research has been recognized with several awards, such as the Dean's Prize for Outstanding Master's and Doctoral Theses and a Best Paper Honorable Mention at SGP. His work bridges computer graphics, computer vision, and machine learning, with a particular focus on extending inverse rendering and differentiable graphics techniques to support a broader range of applications, including content creation and design optimization.

Education

Ph.D. Information Science and Technology – The University of Tokyo	<i>Apr 2023 – Mar 2026</i>
◦ Adviser: Nobuyuki Umetani	
◦ Dean's prize for best Ph.D. thesis	
M.S. Information Science and Technology – The University of Tokyo	<i>Apr 2021 – Mar 2023</i>
◦ Adviser: Nobuyuki Umetani	
◦ Dean's prize for best M.S. thesis	
B.S. Information Science – The University of Tokyo	<i>Apr 2017 – Mar 2021</i>
◦ Thesis Adviser: Takeo Igarashi	

Publications

Inverse Rendering for Modeling with Line Primitives	<i>Dec 2026</i>
◦ Kenji Tojo , Ariel Shamir, Nobuyuki Umetani, Bernd Bickel	
◦ <i>ACM Transactions on Graphics (presented at SIGGRAPH Asia 2026)</i>	
DiffSoup: Direct Differentiable Rasterization of Triangle Soup for Extreme Radiance Field Simplification	<i>Jun 2026</i>
◦ Kenji Tojo , Bernd Bickel, Nobuyuki Umetani	
◦ <i>IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR) 2026</i>	
Strands2Cards: Automatic Generation of Hair Cards from Strands	<i>Dec 2025</i>
◦ Kenji Tojo , Liwen Hu, Nobuyuki Umetani, Hao Li	
◦ <i>SIGGRAPH Asia 2025 Conference Proceedings</i>	
The Mokume Dataset and Inverse Texturing of Solid Wood	<i>Aug 2025</i>
◦ Maria Larsson, Hodaka Yamaguchi, Ehsan Pajouheshgar, I-Chao Shen, Kenji Tojo , Chia-Ming Chang, Lars Hansson, Olof Broman, Takashi Ijiri, Ariel Shamir, Wenzel Jakob, Takeo Igarashi	
◦ <i>ACM Transactions on Graphics (presented at SIGGRAPH 2025)</i>	
GreenCloud: Volumetric Gradient Filtering via Regularized Green's Functions	<i>Jul 2025</i>
◦ Kenji Tojo , Nobuyuki Umetani	
◦ <i>Symposium of Geometry Processing (SGP) 2025 – Best Paper Honorable Mention</i>	
3D Gabor Splatting: Reconstruction of High-frequency Surface Texture using Gabor Noise	<i>Apr 2025</i>
◦ Haato Watanabe, Kenji Tojo , Nobuyuki Umetani	
◦ <i>Eurographics 2025 – Short Papers</i>	
Free-form Floor Plan Design using Differentiable Voronoi Diagram	<i>Oct 2024</i>
◦ Xuanyu Wu, Kenji Tojo , Nobuyuki Umetani	
◦ <i>Pacific Graphics 2024</i>	

Fabricable 3D Wire Art	<i>Aug 2024</i>
◦ Kenji Tojo , Ariel Shamir, Bernd Bickel, Nobuyuki Umetani	
◦ <i>SIGGRAPH 2024 Conference Proceedings</i>	
Stealth Shaper: Reflectivity Optimization as Surface Stylization	<i>Aug 2023</i>
◦ Kenji Tojo , Ariel Shamir, Bernd Bickel, Nobuyuki Umetani	
◦ <i>SIGGRAPH 2023 Conference Proceedings</i>	
Recolorable Posterization of Volumetric Radiance Fields Using Visibility-Weighted Palette Extraction	<i>Jul 2022</i>
◦ Kenji Tojo , Nobuyuki Umetani	
◦ <i>Eurographics Symposium on Rendering (EGSR) 2022</i>	
Neural Motion Compression with Frequency-adaptive Fourier Feature Network	<i>Apr 2022</i>
◦ Kenji Tojo , Nobuyuki Umetani	
◦ <i>Eurographics 2022 – Short Papers</i>	

Work Experience

Postdoctoral Researcher – ETH Zürich, <i>Zürich, Switzerland</i>	<i>May 2026 – Present</i>
◦ Advisor: Bernd Bickel	
Visiting Researcher – ETH Zürich, <i>Zürich, Switzerland</i>	<i>Jun 2025 – Aug 2025</i>
◦ Mentor: Bernd Bickel	
Visiting Researcher – MBZUAI, <i>Abu Dhabi, United Arab Emirates</i>	<i>Sep 2024 – Nov 2024</i>
◦ Mentor: Hao Li	
Visiting Researcher – ISTA, <i>Klosterneuburg, Austria</i>	<i>Dec 2023 – Jan 2024</i>
◦ Mentor: Bernd Bickel	
Software Engineer Intern – Morgenrot Inc., <i>Tokyo, Japan</i>	<i>Mar 2021</i>
Research Assistant – The University of Tokyo, <i>Tokyo, Japan</i>	<i>Feb 2021 – Mar 2021</i>
◦ Mentor: Takeo Igarashi	

Awards

Dean's Prize for Outstanding Doctoral Research – The University of Tokyo	<i>Mar 2026</i>
◦ Awarded for the best Ph.D. thesis of the year in the Creative Informatics Department.	
Best Paper Honorable Mention – SGP 2025	<i>Jul 2025</i>
◦ Recognized for the paper <i>GreenCloud: Volumetric Gradient Filtering via Regularized Green's Functions</i> .	
JSPS Fellowship Supplement for Outstanding Research Progress	<i>Apr 2025 – Mar 2026</i>
◦ Awarded a competitive annual supplement (360,000 JPY) in recognition of exceptional research performance during JSPS Fellowship.	
Japan Society for the Promotion of Science (JSPS) Research Fellowship	<i>Apr 2023 – Mar 2026</i>
◦ Awarded a prestigious national fellowship providing 2,400,000 JPY annually to support doctoral research.	
Dean's Prize for Outstanding Master's Research – The University of Tokyo	<i>Mar 2023</i>
◦ Awarded for the best Master's thesis of the year in the Creative Informatics Department.	

Teaching

CV 802: ADVANCED 3D COMPUTER VISION – MBZUAI	<i>Autumn, 2024</i>
◦ Invited Lecturer – Topic: <i>Differentiable Rendering</i>	

Technical skills

Programming: C++, OpenGL, Eigen, CUDA, Python, PyTorch, pybind11, nanobind etc.

Creative: Adobe Illustrator, Adobe Premiere Pro, Blender, Unreal Engine, etc.